

KETCHUP

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*** START OF THE PROJECT GUTENBERG EBOOK KETCHUP:
METHODS OF MANUFACTURE; MICROSCOPIC EXAMINATION ***

KETCHUP

METHODS OF MANUFACTURE

A. W. BITTING

MICROSCOPIC EXAMINATION

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This brief presentation of some facts concerning the manufacture of ketchup and discussion of the methods for its examination is offered in appreciation for the many favors shown to us by manufacturers. The text has been kept as free from technical terms as the subject would permit, and the results of observations and experiments covered by direct statements instead of giving details and tables.

Nothing new is offered in the method of manufacture, but the doctrine of the use of sound fruit, sanitary methods, and sterilization is reiterated. The position taken upon the method of examination is not new but it is thought proper to present something concerning this phase of the work to the manufacturer.

KETCHUP

Ketchup is a spiced sauce used for its condimental effect in imparting flavor, or to give relish to other foods. It receives its distinctive name from the base used, as, tomato, grape, currant, mushroom, walnut, etc.

The terms ketchup, catchup, and catsup are used to designate any spiced sauce and seemingly without any reason for the one used other than personal preference. Though the derivation of the term has been attributed to different sources by the dictionaries, there seems to be more reason for the use of the term ketchup than for the others, both upon the ground of its prior and more general use, and from the history of its derivation. Murray^[1] gives the derivation of ketchup from the Amoy dialect of the Chinese, the term being **koechiap** or **ke-tsiap**, meaning a brine of pickled fish or shell fish; and he states that the Malayan **kechap**, which has been claimed as the original source, may be from the Chinese, but that the word **kitjap**, as given by some dictionaries from the Japanese, is an impossible word for that language, and is possibly an error for Javanese. The term catchup given by some dictionaries appears to be based on the assumption that the first syllable ketch is a colloquial form of catch. Many manufacturers use the word catsup, a spelling for which there seems to be no etymological warrant. The earliest use of the term catsup, found by the writer, with any particular significance attached to it as distinct from the other two terms, is by Kitchiner, an English physician, in the *Cook's Oracle*, in which directions are given for reducing "catchup" to half the quantity, the statement being that "it may then be called double cat-sup or dog-sup." The first edition of the book appeared in 1817 in England.

1. Murray, J. A. H. *New English Dictionary*.

THE MANUFACTURE OF TOMATO KETCHUP

It is but natural that a product of this kind should vary greatly in flavor due to the selection and quantity of spices, salt, sugar, and vinegar used, and in consistency due to the degree of concentration and fineness with which the base has been comminuted. Most of the recipes for home-made ketchup call for rather liberal spicing and long cooking so that they have a fairly heavy body. These insure good keeping quality, but impart a dark color to the product.

The manufacture of ketchup upon a large commercial scale is of rather recent development and is confined almost wholly to the use of tomatoes as a base. There was little ketchup of the kind best known at present made prior to 1890, as most ketchup was made by what was known as the natural fermentation method, that is, allowing the tomato pulp to ferment spontaneously and using the solid portion for stock. This method was continued, though on a decreasing scale, until 1908, at which time it was practically prohibited. Beginning about 1890, ketchup was made from fresh pulp and barrel stock without fermentation, the fermentation being prevented by the use of a preservative. The method is still in use. The first extensive manufacture of non-preservative ketchup began about 1908, though a few firms had been making it prior to that time, the pioneer probably being E. C. Hazard, of Shrewsbury, New Jersey.

From the amount of space given to the subject of ketchup in the canning and food journals, one might conclude that it is a difficult product to make, or that it is one of very great importance. It is in reality very easy to produce, but has assumed a prominence among food subjects which it does not deserve, due to the fact that some manufacturers have not yet learned the necessity for using care, or persist in using material of questionable quality.

Ketchup is made in the home with very simple apparatus; a colander or sieve for breaking and straining the pulp, and a copper, porcelain, or earthen kettle for cooking, being all that is necessary. The cooking of the tomatoes with the spices, sugar, vinegar, etc., is generally done slowly, until a heavy body is obtained, which results in a dark color, but insures sterility of the product when it goes into the container, and also contributes to keeping quality after it is opened. In the factory many refinements are necessary to make a commercial article which will attract the eye as well as satisfy the sense of taste. The usual dark colored, rough, home-made article will not command a sale over a grocer's counter alongside of that made in a modern commercial kitchen. Here, sorting tables, washing machine, scalding, cyclone for pulping, steam-jacketed kettle, tanks with coils, or vacuum pan for cooking, finishing machine, bottle washing, and filling machine, are all necessary. The pipes carrying the pulp from one machine or vat to another must be enameled, bronze, tin-lined, or silver-plated to prevent the fruit juice from coming in contact with iron or anything which will cause discoloration. The work is done speedily, and the cooking done in the shortest possible time in order to secure the brightest color and smoothest consistency.

The stock should be whole, sound, ripe tomatoes, preferably grown near the factory so that they may be delivered promptly after picking and with the minimum injury. They should be picked when in prime vine-ripened condition. Fruit picked when just turning and allowed to stand one or two days to color will not have the same rich flavor as when vine-ripened, but will stand rougher handling. Green fruit gives a weak color, and over-ripe fruit is prone to become injured and spoil in handling. The tomato should be through the process of manufacture within twenty-four hours from the time that it is taken from the vine. Repeated experiments have shown that rapid handling of fruits and vegetables gives the best results for canning, and the tomato is no exception to the rule when used for ketchup.

The variety of tomato used is of importance. The tomato will vary in solids from less than 5.5 per cent to nearly 8.75 per cent; in soluble solids from less than 3.5 to nearly 6.5 per cent; in sugar from about 2.25 per cent to 4.25 per cent; and in acidity from .3 per cent to .6 per cent. The colors will vary from an almost creamy white to a very deep red with variations in yellow and purple. The only way to get uniformity in a product is to select

one good variety and discard others. The preference is for a clear red smooth tomato of medium size, firm, and of fair acidity. While color may be only “skin deep” as far as being red, yellow, or purple is concerned, experience has shown that a clear red variety gives a better and more lasting color than yellow or purple. A medium sized smooth tomato is preferred because of less adherence of dirt, fewer cracks, and generally more even ripening to the stem. A fairly acid tomato imparts more flavor and needs less vinegar in the finished product. The fleshy portion of the tomato gives the body, but the pulp about the seeds furnishes the characteristic flavor.

The collecting of tomatoes in the field should be done at short intervals so that the fruit may be taken when in prime condition. Where picking is done at too wide intervals, there is a tendency to take fruit that is only colored and not really ripe, and for some to be left and become over-ripe. In both cases the manufacturer is the one to suffer, by increasing the expense of sorting, holding the green if he is to make a high grade product, and by waste from cracking and mashing of the over-ripe. The stems should be left in the field, as they increase the weight and may injure the product to a certain extent.

The handling should be in shallow crates. These should have strong cleats across the ends so that one may be placed above the other without touching the fruit, and if of considerable length, should have a partition. The cleats permit space for ventilation in case they must be stacked for a few hours or more. The depth should be such as not to permit more than three or four layers of fruit. The deep box and the conical basket are not well suited as carriers and should not be used unless delivery can be made by wagon direct from the field and within a few hours after gathering. It is the rule to see cars and barges loaded with baskets arrive at the factory with more or less of the fruit in bad condition. When one basket is set on the edges of two or three others in stacking, there is always cutting of a few of the top fruit, the movement in riding causes others to gradually settle and pack into the cone shape of the bottom, so that if they be held for a day or more, there will be loss of juice, consequent growth of mold, and contamination of the sound fruit from the infected. The actual loss from this form of handling has not been determined, but is undoubtedly much larger than is generally supposed. It is the belief of the writer that the loss is not far from 10 per cent. It is certainly much greater than the difference in the

cost of freight and handling of the box over the basket. All baskets and boxes become more or less infected with mold during the season and this spreads to the fruit, the contamination increasing the longer the fruit is held, the tighter it becomes wedged together, or the greater the cracking. The shallow crate affords the better protection.

When tomatoes arrive at the factory, they should be purchased by weight for **sound** fruit. Buying by the box or basket is antiquated and not satisfactory to either buyer or seller. Under the recent Federal net weight law, purchase by basket or crate must show on each container the exact weight or measure if there be interstate shipment, and the same is true for some of the states. It should not be necessary to give more than a general inspection at the factory. A contract for ripe fruit at ten dollars per ton, which, when delivered, requires sorting, and the holding of unripe and the discarding of defective fruit, is equivalent to ten dollars, plus all the additional cost in labor and the loss in making it fit for use.

If it be necessary to hold the tomatoes for some time at the factory before manufacture, the crates should be stacked in tiers with a foot or more of space between each tier for the circulation of air. Stacking the tomatoes in solid blocks affords the ideal condition for the increase of molds. There can be no doubt that large quantities of fruit have been lost each year from neglect of this simple precaution. Recently a method of holding in water has been originated by Mr. E. W. Grosvenor, at Paoli, Indiana, and consists in using large tanks capable of receiving 500 or more bushels in which tomatoes are submerged in cold water as soon as received, and then held until they can be used. The device is based upon the theory that the tomato skin is practically impervious to water, also that the molds require air for their development and by submergence in water their activity would be lessened.

These tanks are made with false bottoms to receive the sand and dirt, are provided with jets to supply fresh water and to cause the tomatoes to automatically feed upon the conveyor. The first impression is that the tomatoes are soaking in rather dirty water, but tests show that they absorb very little, if any, water, and examination at every stage shows them to be washed cleaner than by the usual method. The work has not been carried far enough to be conclusive, nor to indicate its limitations.

Experiments made to duplicate the factory conditions, comparing air and water storage for short periods, were decidedly favorable to the latter. Much less change occurred in water storage for twenty-four to forty-eight hours than in the air, and there was the further advantage that the tomatoes were washed freer from dirt, sand, and mold, and that rot was cut out better under the water sprays. Some lots of tomatoes were held as long as eighty hours, but this is not to be recommended. When rotting does occur under water, it is of a different character from that in the open and is far more offensive.

If tomatoes be accepted at the factory in a mixed condition, that is, greenish, ripe, and over-ripe, they should be passed first over a sorting belt and preferably one which will turn all sides of the fruit to the inspectors. The green fruit should be held out in separate crates to ripen, and the unfit fruit be discarded. If green fruit be not accepted, the inspection can be done better after washing. In any event the fruit must pass slowly on the table and in single layers. No inspection can be made adequate if the tomatoes pile on the belt two or three layers deep, or pass at such a rate that the eyes tire and all look alike. This is a place where more belts moving slowly, and fewer persons working on each belt, will give the better results. Hand sorting is essential and far more important than in tomato canning. In the latter the defective parts are cut away, but no machine has yet been devised to make the separation complete in making pulp or ketchup.

One other point in inspection is the removal of the stems, which should be the duty of the pickers, but which is often neglected. If the ketchup is to have the brightest, cleanest color, the removal of the stem is advantageous and, furthermore, if the tomatoes are raised on sandy ground, there may be enough sand held around the stem to make appreciable grit. Some manufacturers leave the stem on to give flavor.

WASHING.

The washing is the most important mechanical operation in making pulp or ketchup in order to get a clean product. It is the weak spot in most factories, but fortunately is the one that can be most easily changed. The ideal washer is one that first receives the tomatoes in a tank, holding them for a sufficient length of time to soak and to loosen the dirt, and then submits all parts to a thorough spray under strong pressure. Most washers do not meet these requirements. In many cases the tomatoes are either not dropped into water, or go in and then out again so quickly that they are only made wet and bright, but not clean, then pass under a few cross-sprays, each of which does not deliver a stream more than an inch or so in width, the total spraying not being active over a space of more than six inches and only from above. Some machines do not actually spray the fruit more than one or two seconds. In some cases, it is not so much the fault of the machine as that of the owner in over-speeding and over-loading it. Most machines use a sufficient volume of water, but not under sufficient pressure, nor over a sufficient area. One of the best washers in use is a slight modification of the cylindrical washer used for removing the lye and peel from peaches. It consists of a cylinder about two feet in diameter and twelve feet long, made of a specially corrugated iron. The corrugations are sharper than the ordinary pressed metal used for building and siding, and in addition they are perforated at frequent intervals. This cylinder is mounted on a slight incline. The tomatoes are fed in at one end and the revolving motion causes their discharge at the other. The effect of the corrugation is to cause each tomato to turn over and over in its course and thus avoid all sliding. A spray pipe passed through the entire length and, when provided with the proper nozzle, insures a thorough washing, the tomatoes being under actual sprays from six to twenty times as long as in many machines that are now in use. The water pressure should not be less than sixty pounds per square inch and is better above one hundred pounds if fine perforations or nozzles be used. In nearly every case it is necessary to augment the natural pressure by an auxiliary pump. The principle of the strong pressure is seen in using a hose without a nozzle to wash a floor and one with a nozzle and strong pressure. In the former case it does not clean, while with

the latter it does and with less water. The washer just described is too vigorous for tomatoes for canning, as the treatment is too rough. If the tomatoes are soft or badly cracked, it causes considerable loss, but not of material that should be used in ketchup. The strong sprays will also cut off adherent mold and soft rot. A thoroughly good washer will do about nine-tenths of the work for the inspectors. During the past season some modifications have been made of this washer in the east. The machine has been enlarged, but better results would be obtained by using a greater number of small ones. Again, some washing machines have been ineffective, not on account of any defect, but because of over-speeding.

The vigor with which the washing is done is always apparent in the finished product. The poor washing usually given to tomatoes for canning, accounts in a measure for the relatively large numbers of organisms found in ketchup made from trimmings.

PULPING.

After washing, the tomatoes may be reduced to a pulp in one of three ways: by running the raw tomatoes directly through a grinder and into the cyclone; by passing the tomatoes through a scalding and into the cyclone; and by turning the tomatoes into jacketed-kettles or tanks and cooking them until soft before running through the cyclone. There is a difference in the product obtained by these methods. The first one gives a somewhat larger yield, as the hard parts are cut and torn so that more will be squeezed through the sieve. The color is generally stronger and inclined to the purple side rather than the yellow. The color, however, does not hold so well when exposed to light. The pulp inclines to froth and there is a marked separation of red pigment on the top. A raw pulp will begin to separate into a clear layer below and solids at the top in about fifteen to twenty minutes after standing in a tank. This is due to the air incorporated in the solids and possibly to difference in specific gravity, and not to fermentation, as frequently alleged. Changes will take place more rapidly in such pulp than in that made from scalded fruit.

There is not a great deal of difference between the second and third methods, the object in both cases being the same. If a long scalding be used, the skins will be loosened and the tissue softened so that it will be easily separated from the green parts, hard cores, or black rot. There will be no acquisition of color from the stems to discolor the ketchup. The loss is a little heavier in scalding heating than where the fruit is cooked in the tanks, but there is the compensation that there is less carrying of hard or objectionable material. A scalding to be effective should be much longer than that used in canning, or a greater volume of steam should be used. The tomatoes should be heated to about 180 deg. F. There is little choice in the two methods, but the preference is with the scalding, both being preferred to the raw ground fruit. A pulp made in this way separates slowly and there will be no material increase in organisms for a rather long time (three or four hours). There is less separation of pigment on cooking and there is a clean look to the tissue under the microscope.

In making pulp it is important that the paddles in the cyclone be held back from the screen and the juice driven through by centrifugal force rather than by hard grinding. When kept well back, the green butts, cores, and tissues which have been hardened by brown mold are carried over the end so that there will be fewer black specks in the finished pulp and it will have a better appearance under the microscope.

The pulp should be conveyed immediately from the cyclone to the cooking kettle, and the next operation begun at once. A storage tank is unnecessary when there is large cooking capacity, and in most cases it is a source of trouble rather than a help. A sample should be taken as soon as the batch is drawn, and the specific gravity determined so that the proper quantity may be used to give a finished product of uniform consistency. Assuming that 500 gallons of pulp will give a normal finished batch, if the tomatoes are watery, it may require 550 gallons or more to give the same result when concentrated. This is easily calculated from the specific gravity so that reasonably uniform results may be obtained. Samples should also be tested for acidity once or twice each day so that the addition of vinegar can be governed accordingly. The concentration of pulp will vary from 40 to 60 per cent depending upon its condition and the weight of body desired.

COOKING.

The cooking is done in copper-jacketed kettles, in glass-lined metal, or in wooden tanks, the tanks being heated with coils. The glass-lined tank has the advantage of very little metal coming in contact with the pulp and can be kept cleaner than wood. A question has been raised regarding the suitability of copper for a cooking utensil, though no positive objection has been made. The vacuum pan is coming into use for concentrating pulp, but has been little used in making the finished ketchup. The jacketed-kettle is used by most manufacturers, though the tank and coil is being adopted by those who wish to make large batches, as it is the more economical. Agitators are no longer used, as by proper handling of the steam and automatic traps, little burning occurs on either kettles or coils. The efficiency of the open tank or kettle is increased by providing a strong exhaust or suction for the air at the back and just above the top of the kettle. A swiftly moving current of air across the top of the kettle will carry off the steam and shorten the time of heating from ten to twenty per cent.

A pulp may be reduced in a vacuum pan in about one-fourth the time necessary in the open kettle and with a marked conservation of color and flavor. The vacuum pan may be used for quick reduction and the finish be made in open kettles in order to apply the heat long enough to spice and to sterilize. There are possibilities along these lines which have not been developed.

The time of cooking a batch of ketchup will depend upon the equipment and the consistency of the finished product. With a good kettle or coil and ample steam-supply a batch should be completed in from thirty-five to forty-five minutes. This gives sufficient time to get the most desirable flavor from the spices and is not so long as to result in discoloration.

SEASONING.

The selection of the spices depends entirely upon the flavor desired. Cinnamon, cassia, cloves, allspice, mace, pepper, paprika, cayenne pepper, mustard, ginger, coriander, bay leaves, caraway and celery seed, are all to be found in the various formulae. Some manufacturers spice lightly in order to retain the maximum of the base flavor, while others go to the opposite extreme on the misguided assumption that they will act as preservatives. The quantity used should be determined by the flavor desired and upon no other consideration. The spices may be used whole, ground, or in some cases as acetic acid or oil extracts. The whole spices are preferred by nearly all the manufacturers of high grade goods. They are more expensive, but give a different flavor from the extracts. The spices are weighed for each batch and are tied in a bag or placed in a wire basket and suspended in the kettle while cooking. Some use very large quantities and cook from only ten to twelve minutes in order to get a distinctive flavor. This is very expensive, as only a small quantity of the flavoring matter is extracted in such a short time. One of the serious objections to the use of the whole spices is that they may darken the ketchup and also cause some discoloration in the neck of the bottle. For that reason, black pepper and allspice in particular are being discarded, and oil of cloves is being used in part for the whole berries. The grade of the spice will also have an effect, the cheap stock being unsuitable for a bright clean product. Small quantities of ground cayenne pepper are used as a substitute for the black pepper.

Acetic acid extracts of some of the spices are being used to a certain extent, but they have a peculiar harsh flavor that makes them undesirable. The oil extracts can be used to only a very limited extent, as they impart a flavor suggestive of the drug store.

One method of making a nearly complete extraction of the spices is to place them in their proper proportion in vinegar a few weeks before the ketchup season begins and then add the spiced vinegar in the proper proportion to each batch. The result is different from that obtained by cooking, and the method is not recommended for first grade goods.

The waste of spices in the usual process of manufacture is indicated by some work done by Mr. H. E. Bishop of the laboratory of the Indiana State Board of Health. He found that in making ketchup, when the boiling was kept up for thirty minutes, that only 27.8 per cent of the oil of cassia, 11.5 of the oil of cloves, and 33.3 per cent of the oil of allspice were extracted. (Unpublished report.)

Paprica rosen, Hungarian, or sweet paprika, is used for coloring purposes, though it parades as a spice. This is a mild variety of *Capsicum annum*, one of the species of the genus *Capsicum*, from which cayenne pepper is made. The variety offered to manufacturers has a more intense red color and much less pungency than the ordinary paprika. This paprika can be obtained as the bright fruit, ground dry, or in oil. In the latter, it is said, that part of the capsin is removed, also that the oil sets the color in inferior material. The oil is of a reddish-yellow color and the large number of globules and irregular masses serve to distinguish it from cayenne pepper. It fulfills the claims of the importers—"coloring the ketchup, not adding materially to the pungency, and coming inside the laws in being one of the regular ingredients." It requires just about sixteen times as much as would be required of ordinary paprika to get the same flavor. Considering the cost, in the relative proportion required, there can be little doubt of its real purpose. It will conceal inferiority to ordinary observation in that it gives a red color where otherwise a muddy color might be present. The color does not have durability, and it is easily recognized under the microscope.

Onions and garlic are added in varying quantities and may or may not be kept in the batch throughout the whole cooking period. Considerable difference in flavor is apparent with the length of time of the cooking. Chili peppers are also used in hot ketchup or cocktails.

Vinegar is added to nearly all ketchup. Formerly the acidity was obtained from the fermentation of the tomatoes and the resultant acid was probably mostly lactic. The flavor was different and not so agreeable. A good cider, grain, or malt vinegar may be used. Most manufacturers prefer to use grain vinegar of ten per cent acidity, as the volume required is less and interferes less with concentration. For real flavor, however, this may not be the best. Lately, glacial acetic acid has been substituted for vinegar, a practice which can not be approved and which ought to be abandoned. Citric acid is also added by some. Vinegar is usually added near the finish of the batch, as

otherwise it attacks the kettle to some extent and a part is driven off in boiling. Experiments made by adding vinegar to pulp and evaporating to fifty per cent of its weight in twenty and forty minutes, respectively, show that in the former case the added acidity was decreased in almost the same proportion as the total evaporation, but in the latter case the acid was not driven off quite so rapidly as the moisture. This does not correspond with views held by chefs, as most of them seem to believe that practically all the vinegar is driven off. In order to obtain the sterilizing effect of boiling in an acid medium, it is advisable to make this addition at least five to ten minutes before the end of the cooking period. In home-made ketchup, vinegar is usually added near, or at, the start, and aids in sterilizing the product, as boiling alone may not, whereas, boiling in the presence of an acid will, sterilize.

Oil is not an essential to ketchup, and while a small quantity is often used to prevent foaming, its use in large quantities is undesirable.

Sugar is added to give the desired flavor. The higher the acidity, whether natural, or acquired by adding vinegar, the greater the quantity of sugar needed. In the high grades of ketchup, granulated sugar only is used, but in the cheaper grades, soft sugar or glucose, may be used, though the latter must be declared on the label. The sugar is usually added when the cooking is about one-half completed. There is an advantage in heating both the sugar and vinegar in a separate kettle and adding them while hot, as it will prevent a check to the cooking and lessen the sticking to the coils or kettle.

Salt is used in small quantity and is added near the close of the cooking process.

The use of flour or starch in any quantity for the purpose of making the body thick or heavy is properly regarded as an adulteration. This is also true of pulp from a foreign source, like pumpkin or apples.

The density of the ketchup is left usually to the judgment of the chef, who depends upon the appearance as it pours from the ladle. A quick test can be made by weighing, as done for pulp, but in this case each manufacturer must determine his own standard. A ketchup having specific gravity of 1.090 is apt to be thin; a satisfactory consistency is usually about 1.120 to 1.140.

As soon as the cooking is completed, the ketchup is run through a finishing machine to remove all hard particles of tomato, bits of spice, etc., and to give smoothness to the product by breaking it up into very small particles. There are two types of finishers, the shaking sieves and the rubbing machines. The former is suitable for thin ketchup. The resultant product gives the best possible appearance under the microscope, the tissue showing whole cells, little tearing, and the minimum amount of debris and mold filaments. The objections to the sieve are that the capacity is small and the waste is comparatively large. The rubbing finisher needs to be very carefully adjusted, otherwise it forces practically everything through in a very finely comminuted state. The cells of the tissues are torn to shreds, their contents discharged, molds are broken into hundreds of fragments, and a ketchup may be made to have the appearance of being made from poor material. The finishers have large capacity and will work on either light or heavy goods, but like the cyclone, must be handled with judgment, not attempting to force the last ounce through the sieve.

BOTTLING.

Only new bottles should be used and these should be thoroughly rinsed before using and preferably with hot water. Since new bottles have no tightly adherent particles on the inside, the use of clear water is sufficient, dependence being placed upon the after process to insure sterilization.

The bottling should be done at as high temperature as is practicable, about 165 to 170 degrees F. If the temperature is higher than this, the possibility of burns in handling is increased, and too much space is left in the neck of the bottle after corking, due to shrinkage of the ketchup on cooling, and if much lower, the expansion in processing causes excessive loosening of caps or corks and breakage. Furthermore, when low temperature is used, it requires a very long time to heat the contents of a bottle in pasteurizing. A ketchup is a very poor conductor of heat and the heavier the body, the longer the time that is required.

The closure may be made with either corks or seals, the recent improvements in the latter making them much safer than they were a few years ago.

PROCESSING.

After the bottles are sealed, they should be given a process to insure sterility, the time being about fifty minutes for half-pints and an hour and fifteen minutes for pints—or sufficient time to insure 190 degrees F. for twenty minutes at the center of the bottle.

This step is omitted by many manufacturers, dependence for sterilization being placed upon washing the bottle and subsequent heating for about twenty minutes. The heating is accomplished by conveying the bottles through a chamber containing numerous steam pipes at high temperature and discharging them at the bottling machine. It is assumed that sterilization of the ketchup has taken place in process of manufacture, and the heat within the bottle will care for any infection which may possibly have taken place at a later time from the cap or cork. The safety of this measure depends upon using a fairly acid ketchup or one with a heavy body. It is a risky procedure for mild or thin ketchup. It is a common occurrence to have the stock keep apparently while in the bottle, but spoil shortly after opening. The spoilage after opening is most often due to forms which have been present since manufacture and only need the presence of air to start growth, and are not due to infection from the air. A ketchup will inhibit the growth of organisms which gain entrance from without, while those which are present but held in abeyance through exclusion of air, will sometimes grow. The writer has samples of ketchup put up in 1906 which apparently are sterile, but which will show spoilage within a few days after opening, though done under sterile conditions, and the spoilage be identical in kind with that observed soon after manufacture. How long these organisms will remain alive is not known. In canning, no foods are considered safe without processing, and the same principle is a good one to follow with ketchup.

Processing may be accomplished in open tanks, in retorts, in specially constructed pasteurizers, such as used in the brewing industry, and in hot chambers, the method is not material, though there may be considerable difference in point of economy.

FACTORY ARRANGEMENTS.

The making of ketchup is simple and the factory arrangement for doing the work should be as compact as possible, so that after the pulp is once heated, there is an advantage in having the various steps follow in succession by gravity rather than be conveyed by pumps, especially in small plants. The piping should be as short and direct as possible. The machinery for filling bottles, corking, etc., leaves much to be desired; as separate units they work fairly well, but there needs to be some method devised for handling the bottles automatically from the time they are placed on the washer until they are labeled, ready for the box. At present the time between turning the crate of tomatoes upon the sorting belt until it is ready for the box is only slightly over two hours. Further improvement will not be so much in shortening the time as in eliminating the hand labor.

The foregoing description applies to the making of unfermented, non-preservative ketchup, made from sound stock and delivered into the bottle. Very little ketchup, comparatively speaking, is sold to the consumer in any package other than the bottle. It can be delivered into the bottle when first made, at less expense for labor, with less fuel, and with distinctly less waste than at any subsequent time. It will have a better color and consistency than if stored in bulk and bottled later. It is, therefore, advisable to bottle as much as possible at the time it is made. Ketchup may be packed in bulk in jugs, tin cans, and in barrels, but not satisfactorily; the jug is a poor package; the enamel may be dissolved off the tin can and pinholes form; and the barrel always gives a poor color and off flavor. The best container for bulk ketchup is the gallon glass bottle.

PULP STOCK.

During the height of the season, it may not be possible to convert all the tomatoes directly into ketchup, in which event the surplus may be made into pulp. The first part of the operation is identical with that already described. The concentration is carried just far enough so that subsequently by slow heating for spicing it will give the proper consistency when made into ketchup. A standard has not been fixed, but tentatively it has been proposed that it be at about a specific gravity of 1.035. The concentration may be carried further and water added at the time of the final cooking, but when this is done, the resultant product does not have the same smooth consistency that is obtained by using the thinner pulp. Heavy pulp is made for the purpose of economizing in cans, but experience has shown that economy does not always follow. The higher the concentration, the higher the acid content, and this may attack the enamel and metal with resulting bitter flavor and frequent pinholes. Some manufacturers who prepare their own pulp carry the concentration between 1.030 and 1.033. The method of obtaining this density is to use flasks graduated to hold 500 or 1000 grams of water at 200 degrees F., fill them with the hot pulp and weigh at once. For each flask there should be a proper counterpoise, and the balance be sensitive and weigh in grams. If the 1000-gram flask be used, the specific gravity will be the same as the weight of the pulp. With a valve funnel the flask may be filled level full and the weight taken in less than thirty seconds. For cold pulp, a similar flask is used, but graduated at 60 degrees F. and after filling, the flask is set in a sling and whirled a few times to free it from bubbles, filled again to the level, and then weighed. For pulp of a specific gravity of less than 1.037, this gives fairly concordant results, but the errors increase rapidly the higher the concentration. The same methods may be employed on ketchup. Recently, W. D. Bigelow has improved the apparatus by using a copper flask and adding a handle by which the flask may be submerged in the kettle to take the sample and thus prevents the entrance of air. The use of flasks of any size is described in Bulletin No. 3, National Canners' Association.

The use of the specific gravity method only partly solves the question of standardization. Two pulps each of 1.035 may vary considerably in what the

chef terms body and there is no method of accurately measuring this factor or expressing it. Pulp made by draining will be lighter in weight with the same body, and that from skins and cores will be rough or have the appearance of separating into small flakes or lumps. The specific gravity bears a close relation to the soluble solids, and as these do not have a constant ratio to the fiber in whole fruit, and as the ratio is further disturbed by drainage and in the use of trimmings, it is obvious that the method will not give an exact standard.

Pulp should be filled into gallon or five gallon cans as hot as possible and sealed at once. The practice followed by some manufacturers is to steam the cans first, then depend upon the heat in the pulp to sterilize. The cans are allowed to stand hot for forty minutes, then cooled. The other practice is to give the hot cans a process of about twenty minutes for gallons, forty minutes for five gallons, and then to cool. Cooling is essential to retain color and flavor, as prolonged heat causes "stack burning," producing a brownish color and a bitter taste. The highest grade pulp can not be held in barrels for the reason that the heat is retained too long. Stack burning will take place in glass if the packages are not allowed to cool well in the air before being stored, though the changes are not so marked as in the tin.

PULP FROM TRIMMINGS.

The losses in stock from canning tomatoes amounts to about forty per cent. This is due to the unbusiness-like attempt to can all kinds—very large, very small, and wrinkled, which can not be peeled with economy—to wasteful methods of peeling, and to excessive draining of fruit from handling in too thick layers. In this waste there is much that has good food value and which might be worked up into pulp or ketchup stock if properly done. In order to do this, the tomatoes should be sorted so that only those which are in perfect condition for canning will go to the peelers. These should be medium sized, firm, evenly ripened all over, and free from wrinkles. Such tomatoes can be peeled at the minimum of expense and loss. The sound tomatoes which are small, excessively large, wrinkled, or with green butts, can go in with whole tomato stock. The loss in peeling will then be small and can advantageously be discarded. If it be decided to use trimmings from the peeling tables, provision must be made for extra washing, as the ordinary washer removes little more than the coarse dirt and particles, is not sufficient for unusual conditions or to remove tightly-adhering material, and, furthermore, rot must be eliminated before the tomatoes go to the peelers. The writer has never seen a group of one hundred, or any number, of peelers who will stop to trim and separate rot from peels and cores. Trimming can be done better by a few when sorting the tomatoes than at any subsequent step. If clean skins and cores can be had from the peeling table, they can be converted into pulp and sold if labeled properly, “from trimmings.” Whether such waste is suitable for a good product depends upon how it is handled. For the most part, it has not been handled as well as it should be.

The finished pulp made from skins and cores is not the same as that from whole stock. It contains more fiber, remains more or less lumpy, and lacks the smooth body of whole pulp. The color is not so good, and the flavor is likely to be somewhat different. The flavor of the seed cells and that of the fleshy portion of the tomato are different. Pulp made from each part separately shows marked difference, that from the seed cells being poor in color, but with the more characteristic fruit flavor. Tests show that neither part has any true jellying powers, but that the part from the seed cells gives

the quality of smoothness, the holding together of the particles of solids.
Neither gives a first class pulp alone.

COLOR.

Home-made ketchup generally has a rather dark reddish or brownish color, due to prolonged heating, made necessary under kitchen conditions. At one time this was thought desirable and some of the older recipes call for the use of caramel in order to imitate this color. Most manufacturers now aim to secure a clean, clear color, preferably bright red. This may be obtained when good fruit is used and handled quickly; a muddy brownish or yellowish color is looked upon with suspicion as indicating poor material or defective methods.

The necessity for a clear red variety has already been pointed out, for without proper stock, a superior product of uniform quality can not be made. The tomatoes must be well vine-ripened, as the presence of green fruit and green butts has a decidedly dulling effect. Colorimeter tests show that the use of even small quantities of green material have an immediate dulling effect. Promptness in handling the fruit after the tissue is once exposed to the air is also essential. The tomato, like some other fruits, turns brownish when the surface is cut or exposed. This does not occur as rapidly, nor is it so marked as in apples or in pears, but it is present. When the tomato is converted into pulp, every particle is exposed to the air for a very short time—long enough to make some slight change. The change is most marked in pulp from raw stock and least in that which has been well heated. It naturally follows that ketchup made promptly from whole stock will have the best color, that from canned tomatoes next, then canned pulp, and lastly, that from trimming stock. Pulp allowed to stand hot for too long a time will have a brownish color like stack burning. When barrel pulp was used, this was ascribed to the tannin extracted from the oak.

Pulp should not come in contact with iron at any stage, as the union of the acid of the fruit with the metal will cause discoloration. When such discoloration does occur, it becomes uniform throughout the mass, and not in the neck of the bottle as has sometimes been described.

Darkening in the neck of the bottle is frequently due to the spices used, as has already been pointed out. It can be redistributed throughout the whole by placing the bottle in a shaker for a short time.

Darkening at the top may sometimes be due to extraction of color from the corks. Soaking corks in two per cent acetic acid, then in hot water before drying, and paraffining, will assist in preventing discoloration on cheap grades.

Discoloration in the neck also results from the small amount of air incorporated and from any subsequent addition which may come in through the cork or seal. Bottles which are full to the cork may show no darkening, those having a space of an inch or more between the contents and cork may show little discoloration, while those having more space will show much more marked discoloration. This holds for both pulp and ketchup and in this case the discoloration begins on the surface and works downward. The product made from some fruit will discolor more than that made from fruit grown in another section of the country.

A bright red color is secured in some brands of ketchup by means of paprika, as indicated under spicing.

A light colored ring in the bottom of a bottle is generally due to organisms and debris, indicative of the use of barrel or trimming-stock pulp, or it may result from changes after the process of manufacture. It has been mistaken for sand.

KEEPING QUALITY.

Ketchup must not only keep while in the unopened bottle, but for a reasonable time after opening, if it is to be a commercial success. Every canner understands that if he puts food in a hermetically sealed package and sterilizes by heat, that it will keep until opened. The same principle applies to ketchup in the bottle, but there are some packers who wish to be spared this expense and trouble and prefer to use a substitute for heating.

The keeping quality after opening depends upon the utilization of the same principles followed in the household operation of making fruit butters, ketchup, preserves, and pickles, that is, sufficient concentration and the use of sugar and vinegar. A ketchup can be made essentially a pickle with an excessive quantity of vinegar and it will keep; it can be made a preserve with excess of sugar and it will keep; or, it can be made a distinctive sauce well concentrated in which the vinegar and sugar are used only in sufficient quantity to give proper flavor, and it will keep. Apple juice or cider will spoil quickly if allowed to stand in a warm place; apple sauce will behave in like manner only a little more slowly; but if the juice and sauces be boiled together until they have acquired the consistency or state known as apple butter, they will keep very well. The acidity, sugars, and solids have been increased by the concentration. In the making of tomato ketchup, the fruit does not have sufficient acidity and sugar of itself to give preservative property at the concentration desired for a sauce, so these are augmented by the addition of vinegar and sugar.

A great deal of stress has also been placed upon the effect of the spices in acting as preservatives. Experiments have demonstrated conclusively that when these are used in the small quantities required for flavoring, that their effect is practically nil. The active principles of the spices are effective only when present in the proportion of 1 to 500 or 600 and in ketchup the proportion is only 1 to several thousand. Likewise the quantity of salt is too small to have effect.

The keeping qualities of a mild ketchup will depend far more upon the sterilization than most manufacturers realize. It is easy to make almost any ketchup apparently keep while the bottle is unopened. The spoilage after

opening is most often observed to be due to mold which has been assumed to come from infection from the air. As a matter of fact, this is nearly always due to spores which have been held in abeyance, due to lack of air while in the bottle, and which begin growth as soon as conditions are favorable. Spores which fall into the bottle from the air might be unable to germinate upon such a medium, while those already present would.

CHARACTERISTICS OF COMMERCIAL KETCHUP.

While tomato ketchup is a complex and variable product, its general composition may be determined with a fair degree of accuracy. Inspection will give a good idea of color, consistency, smoothness of body, fineness of finish, tendency to separate, presence of objectionable particles, and evidence of gross fermentation. The odor and taste will give a clue to the kind and quantity of spices used and to a certain extent the character of the raw material. Judging by odor and taste is not so well done as judging by the eye by most persons. The education of those two senses has been neglected and therefore fail to give all the information which might be acquired in this way.

A chemical examination which will give the specific gravity, total and soluble solids, sugar, salt, and total and volatile acidity, will be sufficient to give a good idea of the stock used—tomato, salt, sugar, and vinegar, but not the spices. A microscopic examination will assist in determining the condition of the material used and whether decomposition has taken place before or after manufacture. The facts obtained through these sources will permit of classifying commercial ketchup with a fair degree of accuracy.

There has been a very marked change in the character of ketchup since the transition from the preservative to non-preservative goods, not only microscopically, but also in composition. Formerly, there were very many brands of thin liquid ketchup, showing little concentration of pulp, very low in sugar, and having only small quantities of vinegar; the standard was bulk rather than quality. The microscopic examination also showed that the product had frequently undergone change before and after preparation. Recent examinations show that there has been a very marked improvement; that the body is decidedly heavier, more sugar and vinegar are used, the tissue is cleaner, and there are fewer organisms present, also that the difference in composition in preservative and non-preservative ketchup is small, whereas, formerly it was marked.

The variations found in ketchup of rather recent examination show in the non-preservative kind the specific gravity varied between 1.091 and 1.177; the solids between 19 and 37 per cent; the salt between 2 and 4 per cent;

sugar between 12 and 29 per cent; and volatile acids between .54 and 1.24 per cent. In the preservative kind, the specific gravity ranged from 1.032 to 1.120; the solids from 9.23 to 28 per cent; salt, 1.48 to 3.4 per cent; sugar, 4.95 to 16.9 per cent; and volatile acidity, .16 to .64 per cent. As a class they averaged lower in concentration of tomato and in sugar and vinegar, though if proper sterilization had been used, some of them would have kept without difficulty. In experimental work it was found that a ketchup concentrated so that when finished it showed an added sugar content of 15 per cent or more, a total acidity of 1.2 per cent, and a specific gravity of 1.120 or more, that it would keep. To obtain a total acidity of 1.2 per cent means the addition of about .4 to .6 per cent acidity in the vinegar used. However, there are brands of ketchup on the market which keep well after being opened and which have a total acidity of less than 1.0 per cent.

The manufacturer can use the following as a starting point for non-preservative ketchup; pulp, 100 gallons; sugar, 60 pounds; salt, 8 pounds; vinegar, 100 grain, 2 gallons; spice to flavor; and concentrate to 50 to 55 gallons.

MICROSCOPIC EXAMINATION.

A discussion of the microscopic appearance of ketchup in terms which can be readily understood by manufacturers is not an easy task, as it necessarily involves technical knowledge. The subject has become one of importance, owing to the attitude of many food officials in enforcing a microscopic standard for this product, and on the part of many brokers in requiring a guarantee to comply with this standard in making purchases. Many manufacturers have either assumed or found it necessary to have their finished products examined. Some employ "experts" to make the examinations in their own plants, while the majority send their samples to commercial laboratories. The total tax upon the industry for such work amounts to thousands of dollars annually. The result of the work as a whole has been beneficial, as any effort is which attracts attention to details. It has likewise been the means of causing much unpleasantness and not infrequently loss, because of lack of understanding on the part of both manufacturer and examiner as to the cause of certain findings. The manufacturers have proceeded in the usual way without sufficient knowledge of what the resultant product will be unless there is careful supervision of material and methods, while too frequently the examiner is neither experienced in technique of the examination nor in the effects of the different steps in manufacture upon the product. Furthermore, much distrust in microscopic finding is evinced when a half dozen or more samples from the same batch, sent to as many persons, result in as many different reports. It naturally causes a lack of confidence in both paid examiners and in food officials, though those who make these examinations may be absolutely honest in their findings. In order to clarify some of the points, it has become necessary to go into detail, into both the method of examination and into the effect produced by manufacture.

A scientific method of food examination is necessary for food officials in order to determine the condition of a product, but is not necessary for the

manufacturer, though it may be advantageous. The latter is in a position to know what enters his factory and what changes take place in the food until it reaches the sealed package. He should have no fear of a method which correlates the findings in the finished product with that of the material used and the changes due to treatment.

Undue importance may seemingly be given to the subject of ketchup, but the principle involved applies as well to other products.

The fundamental basis for the microscopic examination of any food product must depend upon the structure of the material which enters into its composition. Any attempt to determine an abnormal condition, such as decomposition, without a knowledge of the normal, must necessarily be of little value. There is some work which can be done in a mechanical manner by almost anyone capable of looking through a microscope, and if the work is properly supervised, it may have a value, but the lines along which this can be done are very limited. Any attempt to apply such superficial methods to the general examination of food products can not properly protect the public and may be unfair to the producer. It has, therefore, been deemed advisable to incorporate a brief statement concerning the structure of the tomato before discussing the resultant products.

HISTOLOGY OF THE TOMATO AND OF THE RESULTING KETCHUP.

STRUCTURE OF THE TOMATO.

Pericarp. The tomato is a typical berry, the ovary wall, free from the calyx, forming the fleshy pericarp, which encloses chambers filled with a clear matrix, containing the seeds. The pericarp consists of an outer tough membrane, the epidermis, a more or less thick layer of parenchyma tissue, the pulp, and an inner thin, delicate membrane, the lining layer of the loculi or chambers in which are the seeds. The epidermis consists of a single layer of cells which have a very thick continuous cuticle about one-half of the diameter of the whole cell. The cuticle differs in chemical composition from the rest of the cell walls, being impervious to water, and resisting rotting longer than do the cellulose walls. As it is continuous over the whole of the fruit, the skin can be readily separated from the other tissues. Hot water facilitates the removal of the skin, as it causes the cellulose of the walls to swell more than the cuticle, producing an effect as of shrinkage of the outer wall and a consequent curling of the skin. The radial walls of the epidermis are short and irregularly thickened, leaving pits in the walls, and giving them a beaded appearance. The skin constitutes about 1.3 per cent of the tomato.

The layers of parenchyma just beneath the epidermis are closely united and flattened, with their adjoining walls irregularly thickened. On account of their position, they are called hypoderm. In the tomato the hypoderm consists of two or three layers of cells, parts of which usually separate with the epidermis. Below these cells are the thin-walled parenchyma cells, which are approximately globular, vary considerably in size, are very loosely held together, and have many intercellular spaces. These cells constitute the mass of the pulp, and with the juice constitute 96.2 per cent of the tomato.

The layer of cells which lines the chambers has the typical leaf epidermal structure, the wavy outlines, the hollows and protuberances of adjoining cells fitting one another so that they form a continuous layer. They are also

flattened laterally. The structure can be understood readily when it is known that the pericarp is really a metamorphosed leaf and that the outer side of the leaf forms the inner wall of the ovary.

The chambers of the tomato are filled with a clear, slimy matrix in which the seeds are embedded. The matrix consists of parenchyma cells of various sizes and with delicate walls, and a small nucleus. The cells are massed loosely, and can be separated readily. In those cells, as well as in the wall cells, are starch grains which vary in size, being round or approximately so, and having the hilum, when visible, a straight line to one side of the center.

Coloring Matters. In the parenchyma cells are two coloring matters, one yellow, which is amorphous in structure, and the other red and of crystalline form. The sap contains a yellow color in solution which differs in its reactions from those in the pulp.

Red Color in Tomatoes. The red coloring matter in tomatoes is in the form of irregularly shaped crystal-like chromoplasts, which occur in masses of various sizes. They are present in largest amounts usually in the protoplasm which lies close to the ectoplasm and in that surrounding the nucleus. They vary from sharp, bright-colored forms to those more or less blunt in outline, and dull in color. They may be situated largely in the periderm, the soft parenchyma beneath the periderm, or through the whole mass of the parenchyma with the exception of the matrix surrounding the seeds in the loculi. In tomatoes having the color in the periderm a considerable amount is lost by adherence to the skin. The chromoplasts are not affected by rotting to the same extent as are the other constituents of the cell; they can be found floating free in the debris from rotted cells, still retaining considerable color. They lose their color gradually, in some varieties much more rapidly than in others. In stored pulp which has fermented, the color may be faded to a dull yellowish brown. In tomatoes intended for ketchup where a bright red color is desirable, care should be used in the selection of a variety having the chromoplasts bright, properly oriented, and in sufficient quantity.

Vascular Bundles. In the pulp of the tomato are found strands of vascular tissue, entering from the stem, and dividing and ramifying through the soft pulp. These consist of long tubes with thin walls, some of which have a strengthening band in spiral form on their interior walls, the

associated cells being without any special marking. The strands vary in size from those having a few tubes to those having a large number.

Seeds. The seeds of the tomato are small, flattened, yellow bodies covered by a clear gelatinous membrane. Their peculiar characteristic is the out-growth of hairs of varying lengths. The seeds constitute about 2.5 per cent of the weight of the tomato.

STRUCTURE OF KETCHUP.

Although the tomato pulp is broken into fine particles by the action of the cyclone, and the skin and seeds are removed by the fine sieves, pieces of the various tissues can be readily identified. The skin and seeds have characteristics which would serve to distinguish them from similar parts of other vegetables which might be used for adulteration, but particles of skin and hairs from the seeds are rarely found. The distinctive features which can be relied upon are the red, irregularly-shaped, chromoplastic bodies in the parenchyma cells, and the peculiar wavy-outlined cells of the lining layer of the chambers. As nearly all young vegetable tissues have spiral vessels in their vascular strands, these are not distinctive, except that they might differentiate similar tissues of different size. There is very little starch in mature tomatoes, and moreover, as the cooking causes the starch to swell and lose its structure, the starch could not be used for identification.

Good ketchup made from whole tomatoes, in spite of the minuteness of the particles, has a clean appearance, and can be readily distinguished from poor ketchup. All ketchup will have some micro-organisms present, as it is practically impossible to free the tomatoes from them in the washing, but the number is very small in some of the best, in the manufacture of which careful washing and sorting have been done. The poorer the ketchup, usually, the greater number of organisms—bacteria, yeasts, and molds; sometimes one form predominating, sometimes all three being in great abundance, this latter condition usually prevailing in the poorest ketchup, where more or less rotting has occurred.

As the tomato pulp is a favorable medium for certain organisms, these will develop first, and it has also been determined that while one organism is developing vigorously, others present are checked until the activity of the first ceases. Then again, as the composition of the pulp is being altered by the development of the organisms, the changes induced render it a more suitable medium for other organisms which are present but held in abeyance, so that pulp which has been allowed to stand for some time will usually have present not only a large number, but also different kinds of organisms.

CHANGES PRODUCED IN PULP BY ROTTING.

When tissue is held and allowed to rot spontaneously, the pulp is decomposed into a granular, watery mass. The cells beneath the epidermis are the finest and driest in the sound tomato, considerable pressure of the cover-glass being required to separate them for examination. Even when forced apart, the cells retain their shape. They contain a delicate semi-transparent protoplasm with a rather large nucleus surrounded by protoplasm and having strands from this mass connect with the protoplasm lining the wall. Pieces of the same tissue, on having the skin removed so as to expose the broken tissue to the air, were covered with mold in one day and in three days so badly disorganized that the cells separated with the weight of the cover-glass. The cells were transparent, the walls collapsed into a wrinkled mass, the protoplasm had disappeared, except a skeleton of the nucleus, but the red chromoplastic masses were intact. The middle lamella of the cells is the part which dissolves first, allowing the cells to separate and causing the walls to become thinner. The cell cavity is often filled with bacteria, so that the effect of the rotting can not be seen until the cells have been washed thoroughly. These bacteria have been mistaken for the particles left by the decomposition of the cell contents. The vascular bundles are surrounded usually by small parenchyma cells which do not separate readily from the strand in the healthy tissue, but in the decayed tissue the vessels can be seen clearly, free from other tissue. In advanced stages of rottenness the walls of the vessels may be dissolved, leaving only the spiral thickening, and the parenchyma tissue crumbled into powder-like fragments. The parts of the tomato which resist rotting the longest are the skin, which may be washed clean of adhering particles, the spirals of the vessels, and red particles of the chromoplasts.

The conditions found in the rotted sections and pieces of tomato can be distinguished in the poor ketchup and these factors, together with the large number of organisms present, serve for purposes of differentiation.

ORGANISMS IN KETCHUP.

Tomato pulp furnishes a medium suitable for the development of many organisms, as it contains all of the necessary food elements. The raw pulp has an acidity of from 0.2 to 0.4 per cent usually, though there may be variation due to fermentation and other causes. On account of its mild acidity, it is especially suitable for the development of many yeasts and molds, and some forms of bacteria, consequently there is present a varied and abundant flora if the pulp be held for an appreciable time before using, or if it has been made from tomatoes not properly sorted and washed. Where the black rot occurs on tomatoes, the tissue is hardened like cork, and if not removed on the sorting belt, is broken into small pieces by the cyclone, and appears as black specks in the ketchup, these being readily perceived by the naked eye. The white rot forms soft spots, which, though not so prominent as the black, carry much more contamination, as, apart from the bacteria, yeasts, and molds present, they are often swarming with Protozoa. These are not ordinarily recognized in the ketchup, as a chemical or physical shock causes them to contract, assume a spherical shape, and become motionless. In this condition they resemble the immature conidia of some of the molds. Rarely only one organism predominates in pulp from rotted fruit, then the rot consisting of a nearly pure culture. In all cases of soft rot, there is much more contamination carried, as the organisms are small and a greater number present in a given area. Whenever the inner tissue of tomatoes is exposed, organisms develop rapidly, the forms varying with the locality and the conditions in the pulp. Some of these organisms may survive the treatment of the pulp when converted into ketchup, or the original organisms may be destroyed, and a different set gain access and develop, but in either event all the organisms alive or dead which were present at the period of manufacture are found in the ketchup. It has been noted that certain brands of ketchup have predominating organisms present which are practically constant from year to year.

A method for the microscopic examination of ketchup in order to determine the number of organisms present is described in Circular No. 68, Bureau of Chemistry. It consists in an adaptation of a method used in examining blood in physiological and pathological work, and of yeast in the

brewing, wine-making, and distilling industries. The outfit required consists of two parts, the microscope and the counting chamber, each with minor accessories. The optical outfit recommended for food examination consists of a microscope with eye pieces and objectives which will give approximate magnifications of 90, 180, and 500 diameters. It is advised that these magnifications be obtained by using 16 mm and 8 mm apochromatic objectives, and $\times 6$ and $\times 18$ compensating oculars ($\times 6$ ocular and 16 mm objective equals $\times 90$; $\times 6$ ocular and 8 mm objective equals $\times 180$; and $\times 18$ ocular and 8 mm objective equals $\times 500$), higher objectives being impracticable on account of their short working distances. This equipment is adequate for working upon blood or yeast, but is wholly inadequate for bacteriological work, except that of the simplest character and under conditions quite different from those found in ketchup and other food products.

The counting apparatus or chamber recommended is known as the Thoma-Zeiss haemocytometer, named from the designer and maker. The apparatus consists of a heavy glass slip, on which is cemented a glass 0.2 mm thick, having a circular hole in the middle. In the center of the hole is mounted a smaller disk 0.1 mm thick, leaving an annular space. In the middle of the small inner disk are etched two sets of twenty-one parallel lines which cut each other at right angles. The drop of liquid to be examined is placed on this square, after which it is covered with a specially heavy cover-glass, which, if perfect and adjusted so closely that Newton's rings appear, gives a layer of liquid 0.1 mm in depth. The drop to be examined must be so small that it remains in the middle of the chamber, but in contact with the cover-glass and bottom of the cell. Each side of the ruled square is 0.1 mm, and as there are 20 spaces on a side, there is a total of 400 small squares, the depth being 0.1 mm, thus the cubical content of each is 1-4,000 c mm or 1-4,000,000 cc. For convenience in counting, every fifth space is sub-divided. Other counting chambers have been devised based on the same principle, but varying chiefly in their rulings for convenience in counting.

The other apparatus recommended consists of a 50 cc graduated cylinder, slides, and cover-glasses.

Since the counting chamber has been used extensively in blood examination and in yeast work, a brief description of the technique as followed in the latter may serve to give a better understanding of its

limitations. First, in the preparation of the sample, the cylinder and flasks for mixing, and the pipette must be absolutely clean. The liquid to be examined is shaken thoroughly and then the measured sample withdrawn as quickly as possible to prevent the cells from settling and diluted with weak sulphuric acid (about 10 per cent), which prevents any further development of cells, and also aids both in the separation of the cells from one another and in their suspension—the latter factor being important when only a single drop is taken for examination. When counting blood cells, a normal or other salt solution is used so as to have the specific gravity of the diluent approximately that of the blood serum. The dilution is made as low as possible, since the number obtained in the count has to be multiplied by the dilution co-efficient, and any errors made are increased proportionately. A slight error when multiplied by the factor 4,000,000, the unit for each square, becomes very large in the total. The sample is shaken very thoroughly after the diluent is added, a drop of the liquid taken by means of a pipette, placed in the center of the counting chamber, and the cover-glass put in place. The withdrawal of the pipette and the transference of the drop to the chamber are done as quickly as possible to prevent the cells from sinking. The determination of the number of blood corpuscles, yeasts, or other cells in one cubic centimeter, the unit of volume generally used, will depend upon the average found in a number of squares. The number of squares to be counted is determined by making counts until a constant average is obtained, for if a true average is not obtained, the counting, naturally, is of no value. If the mounts do not show uniformity in the field, they are repeated.

In using the counting chamber for counting yeast cells and blood corpuscles, for which it was originally devised, the bodies to be examined are fairly large, well defined, and suspended in a fairly clear liquid, usually of rather high specific gravity. Even with these favorable conditions, the work must be done by observing the most careful technique in order to get relative results, which will be of value, and they are absolutely useless if any detail has been slighted or neglected. In attempting to adapt the method to food products, very different conditions are encountered—conditions which are opposed to obtaining accurate results. Food products, like ketchup, consist of a mixture of solids and liquids in which are various forms of organisms, the latter in varying condition, due to their environment and treatment, as well as to stages of disorganization.

In estimating the number of yeasts and spores in pulp or ketchup, the Thoma-Zeiss counting chamber is used and the mount observed under a magnification of 180 diameters. To prepare the sample, 10 cc of the material has 20 cc of water added and is “thoroughly mixed.” Before taking a drop for examination, the sample is allowed to rest for a “moment” to allow the “coarsest particles” to settle. This step in the technique is not as clear as could be desired, for what might be considered as “thoroughly mixed” by one microscopist as a half dozen shakings of the cylinder, might not be so construed by another even with sixty shakings. As the material consists of both solids and liquid, this is a very important detail, as it may easily account for some of the wide differences in results obtained by different workers on the same sample. In a bulletin^[2] dealing with the examination of solid foods, the following statement occurs relative to the shaking in order to be able to obtain the bacterial condition: “The longer the shaking, the more perfect was the diffusion of particles. It could not, however, be continued beyond a comparatively short period of time, because of the multiplication of organisms. With the quantities of tissue above stated, ten minutes’ shaking was selected as a happy medium between an undesirable multiplication of the organisms on the one hand and the retention of the organisms by the tissue and the consequent lowering of the numbers found, on the other.” The organisms in pulp or ketchup are dead, or, if alive, do not possess such phenomenal power of multiplication, therefore, the shaking should be conducted with sufficient energy and for a sufficient time to insure their separation from the tissue. Furthermore, “letting stand for a moment” may mean thirty seconds or two or three minutes to different persons.

2. No. 115—Bureau of Chemistry, Dept. of Agr.

In all biological work involving the counting of organisms, either by the plate or direct method, in the case of yeast, the operator works as rapidly as possible to prevent the organisms from settling, so as to have them evenly distributed in order that he may obtain an average sample. A pipette is used for removal of a drop of the liquid and the drop placed in the chamber as quickly as possible to prevent settling. No directions are given as to how the drop of the diluted pulp or ketchup is to be removed to the chamber, so that a stirring rod or other apparatus is frequently used, as the solid particles

interfere with the use of a fine pipette. If the rod be inserted to the bottom, or nearly to the bottom of the mixture and withdrawn slowly and another withdrawn somewhat rapidly, a difference of fifty per cent or even more may result in the count. It is not possible for different operators to use pipettes, glass rods, pen knives, toothpicks, and matches for drawing the samples, and get comparable results. It has been found that in (all of these have been seen in use) the counting of the organisms in pulp and ketchup, some persons use distilled water, others tap water, some clean their measuring flasks and pipettes, while others rinse them, so that naturally reports are made of such varying numbers that manufacturers do not look upon the method with confidence. It is only by using uniform methods and the same care necessary for other biological work that even an approximation can be made.

STRUCTURE OF THE TOMATO.

To obtain the number of yeasts and spores in the sample, a count is made in one-half of the ruled squares. Two hundred squares represent a volume equivalent to 1-20 c mm, which, multiplied by the dilution, would give the number in 1-60 c mm. It is stated that it is believed that it is possible for manufacturers to keep the count below 25 per 1-60 c mm.

The same mount is used in estimating the bacteria, but the $\times 18$ ocular used so as to increase the magnification to approximately 500 diameters. The "number in several areas, each consisting of five of the small squares, is counted." Nothing is said as to the order of the five squares, whether in a row or other arrangement, nor what number constitutes "several." The average number found in five squares represents the number in 1-800,000 part of a cc, and this multiplied by 3, for the dilution, would make the factor 1-2,400,000 for a cc. It is stated that it is believed that it is possible for manufacturers to keep within 12,500,000 bacteria per cc in the pulp and 25,000,000 in ketchup. The number present is expressed in terms per cc though the yeast and spores are expressed in 1-60 c mm. Possibly bacteria to the lay mind mean something dangerous, so by expressing the numbers in millions they appear appalling. Yeasts and spores are not so generally associated with dirt and disease so that by giving them a small unit, only 1-60,000 part of a cc, they may seem much less offensive. If the mind is capable of conceiving what is meant by millions per cc for bacteria in one case, there seems to be no good reason why the same unit of volume should not hold for the other.

To estimate the number of molds present, a drop of the undiluted pulp or ketchup is placed on an ordinary slide and the ordinary cover-glass pressed down until a film of 0.1 mm is obtained. The directions state that after some experience this can be done, but do not state how one's efforts may be directed to obtain this result. It is apparent that by experience in comparing a measured amount with a judged amount that the tendency would be toward accuracy, but in this case there is no measured amount for comparison, except the diluted drop in the counting chamber. Some workers have placed thin cover-glasses under the edges of the mount so as to have

something to help in estimating the thickness of the film, but as the thinnest ordinary cover-glasses vary from .12 to .17 mm in thickness, the error varies 20 to 70 per cent from that required. One manufacturer in advertising No. 1 cover-glasses states that they vary from 0.13 to 0.17 mm, while another states they vary from 1-200 to 1-150 of an inch (0.127 to 0.169 mm). Careful checks show that it is not always easy to get exactly .1 mm on the specially prepared counting chamber; that unless the cover be placed with care and pressed uniformly on all sides until Newton's rings appear, a variation of ten per cent or more in thickness may occur, and without such a guide the error becomes greater. The micrometer screw adjustment on the microscope can be used to help in determining the thickness, but none of the workers observed has used this refinement.

The examination for mold is made with the $\times 6$ ocular and 16 mm objective, giving a magnification of approximately 90 times. About 50 fields are supposed to be examined and the result expressed in terms of the per cent in which mold was found. It is stated that it is believed that manufacturers can conduct their operations so that mold will not be present in more than 25 per cent of the fields. There are, therefore, three units in which to express the results: bacteria in cubic centimeters, yeasts and spores in one-sixtieth of a cubic millimeter, and molds in percentage of microscopic fields.

Aside from the errors which may occur in the manipulation of the purely mechanical part of the technique, there are other considerations which affect the accuracy of the results. First, the differentiation between organisms and tissues is not considered possible by most pathologists and bacteriologists without differential staining. Even in such simple examinations as those for diphtheria and tuberculosis, a stain is required. In foods the particles of the plant tissue and the organisms are not so different that they can be clearly separated without using similar technique. It is possible to make some separation, but not with accuracy. Threads of protoplasm may be mistaken for bacilli; the granular contents of a cell for cocci, yeasts, or spores; bits of cell wall for hyphae under the magnifications given, and the results obtained be high or low, depending upon the personal ability of the operator. Each error magnified by the enormous factor used in calculating the final result naturally gives figures which may be far above or below the truth. Those who have had special

training in plant structure and bacteriology are likely to give the higher figures, while those who have had these subjects as incidentals in a scientific course are apt to give much lower ones.

Second. The standard is set for what organisms shall be counted and those which need not be. It is said that micrococci need not be counted because of the difficulty in distinguishing them from “particles of clay, etc.,” and not upon their power to produce decomposition. When an organism is a coccus and when rod shaped is not easily settled, even with the aid of pure cultures and high power objectives. More than one organism has found a home first in one group and then in the other, and differentiation with the low power obtained by an 8 mm objective is impossible. There are always present some very large rods, but there may be more very short ones which may not be counted, and there is nearly always a diplococcus present, which, with the magnification used, is difficult to differentiate from a rod. There are four forms associated with rot and tomato diseases which have been carefully studied—all rods, but very small ones. *Ps. fluorescence*, $0.68 \times 1.17-1.86$; *Ps. michiganense*, $0.35-0.4 \times 0.8-1.0$; *B. carotovorus*, $0.7-1.0 \times 1.5-5$; and *B. solanacearum*, 0.5×1.5 . *Bacillus subtilis*, $.7 \times 2-8$ and some lactic acid forming varieties are always present. It is clearly a matter of judgment on the part of the examiner as to which organisms he will count and which he will not attempt to count. A personal equation is thus introduced which nullifies the possibilities of scientific accuracy.

The yeasts and spores are counted together. They can not be separated under the microscope, neither can they be differentiated from contracted protozoa which may be present in large numbers. In counting these, it is not always possible to distinguish the smaller yeast cells and smaller spores from the refractive bodies which are formed in some mold hyphae when these are impoverished, and which are liberated if thorough shaking of the sample be done. The yeasts found in pulp and ketchup are more likely to be “wild yeasts” and these are, as a general thing, smaller than the cultivated, sporulate more readily, and have more highly refractive spores. Then, some of the so-called molds found form minute conidia and when these and the yeasts are mixed with the detritus of the tomato and the mass subjected to heat, with the consequent changes, the accuracy of the count becomes a somewhat problematical matter. A careful examination of the kind and

condition of the hyphae present might assist materially in making some distinction.

In counting molds, no distinction is made as to whether a small bit is in the field or a large mass. In making a mount for molds, the solids generally tend to stay in the center of the field while the liquid tends to run to the edge. The fields selected may therefore give a high or low result determined by their location. One examiner desiring to favor the manufacturer may select the outer part for most of the fields, while another, making the examination for the buyer, who may wish to make a rejection, may reverse the operation. Some persons modify the directions given by counting only pieces which are one-sixth the diameter of the field, while others use a smaller fraction. It is easily possible to have one clump of mold in one field which will be twenty to thirty times in extent that of another, yet both are given equal value in the final expression.

Third. No real relation exists between the organisms counted and decomposition, for mere numbers are not always coincident with putrefactive activity. A pulp or ketchup may be bad and show less than 30,000,000 bacteria, or it may be good and show 300,000,000. Rotting, or decomposition, may depend more upon the cocci and the organisms which are not counted than upon those which are. The only work done in which microscopical and chemical work were reported on the same samples appears in Circular No. 78, Bureau of Chemistry. This was not done upon samples prepared and kept under control, but for the most part upon commercial pulp and ketchup. The results do not show any close relation between the number of organisms and the lactic acid content which is given as the measure of decomposition.

Fourth. Bacteria are expressed in numbers per cc, yeast and spores in numbers per 1-60 c mm. Since the counting can be done only in the fluid portion, an error occurs proportional to the number of bacteria in or attached to the tissue which cannot be counted.

The error of assuming that numbers of organisms alone are a sufficient index of the wholesomeness of a food product is well illustrated by work on water analysis. The following statement by an authority on the subject is illuminative: "The belief is widespread among the general public that the sanitary character of a water can be estimated pretty directly by the number

of bacteria it contains. Taken by itself, however, it must be admitted that the number of colonies which develop when a given sample of water is plated affords no sure basis for judging its potability. A pure spring water containing at the outset less than 100 bacteria per cubic centimeter may come to contain tens of thousands per cubic centimeter within twenty-four to forty-eight hours, after standing in a clean glass flask at a fairly low temperature. There is no reason for supposing that the wholesomeness of the water has been impaired in any degree by this multiplication of bacteria.”^[3]

3. Jordan, E. O. A text-book of General Bacteriology. 1908.

There are certain steps in the process of manufacture which also influence the number of organisms which may be counted. A pulp may vary from an unevaporated tomato juice to a concentration which is represented by an evaporation of a volume of water up to 60 per cent, and ketchup may vary from a thin watery consistency to one which is so heavy that it will scarcely flow from the bottle. It becomes evident that a method which does not sustain some close relation to the amount of tomato present would naturally be deficient as a standard for judging. For example, a tomato juice with an initial count of 10,000,000 if evaporated to one-half its volume will have more than twice the number of organisms estimated in the original. The pulp is composed of both liquid and solids and part of the liquid portion only is driven off by evaporation, leaving in the residue a different proportion to the solids. As the organisms can be counted only in the liquid portion, it is obvious that with concentration, the number will be increased at a much greater ratio than will the reduction of the bulk. A thin pulp with 10,000,000 bacteria may easily be worse than a heavier one with 30,000,000 or 40,000,000, if judged by numbers alone. The same conclusion is necessarily true for ketchup. It clearly refutes the argument that a product having twice as many bacteria as another of the same kind is more than twice as bad. The effect of recommending an arbitrary low limit for bacterial content, irrespective of the consistency of the product, is to cause manufacturers to pack thin pulp and sloppy ketchup, and to discourage the more desirable heavy body. The examination of a very large number of samples shows that the majority of the heavy pulps and ketchup

upon the market show much higher counts than the thin ones when the tissues show good stock in both.

It is not possible to concentrate any pulp to the consistency of paste and have it pass under the present method; that is, considering a product to be filthy, putrid or decomposed if the bacteria exceed 25,000,000 per cubic centimeter.

There are some soup and ketchup manufacturers who still follow the draining method for separation and this is generally done to secure a certain quality in the flavor. This kind of pulp always shows a high bacterial count, which is usually ascribed to fermentation. As the draining can be started in about twenty minutes, and is nearly always completed in forty minutes to one hour, there is little time for fermentation, and yet such a pulp may show several times the count of the original whole pulp. The condition is similar to that which takes place in the separation of cream by gravity. Dr. John F. Anderson, U. S. Public Health Service,^[4] has shown that the bacterial content of gravity cream is about sixteen times that of bottom milk and that this discrepancy may be much wider. One test is given in which the cream showed 386 times as many organisms as the bottom milk. The question logically arises whether, if a pulp which contains 10,000,000 bacteria per cubic centimeter and is considered sound, becomes "filthy, putrid or decomposed" when the same pulp is heavily concentrated and the count becomes 100,000,000, or a cream is bad when it contains 2,000,000, though the whole milk from which it was derived contained only 300,000. There should be a recognized difference in rating a product in which the number of organisms is influenced by concentration, and one in which they have developed. Some very erroneous statements have been made upon increase of bacteria in pulp while standing. Some of these have been based upon the academic proposition that reproduction in bacteria may occur every twenty minutes under perfect conditions of food supply, freedom of movement, and optimum temperature. Such statements are obviously not based on experiments with pulp. Assuming that such a rate of reproduction were possible, a pulp with an initial start of only 5,000,000 would increase to 10,000,000 in twenty minutes; 20,000,000 in forty minutes; 40,000,000 in one hour; 80,000,000 in one hour and twenty minutes; 160,000,000 in one hour and forty minutes; 320,000,000 in two hours; and 2,560,000,000 in three hours. No food product like tomato pulp, cider, or grape juice would

be usable in a very short time. To determine the rate of increase of the organisms in tomato pulp, experiments were made, using sound tomatoes. In each experiment, the tomatoes were divided into two lots, one lot used raw, the other steamed, the steaming varying from two minutes' time, just sufficient to slip the skins, and eight minutes, in which the whole tomato is softened. Samples were taken at hourly intervals for the first six hours, then at intervals of twelve hours, the samples counted by means of the plate and direct methods. For the plates tomato gelatin was used with an acidity of 0.3% and 0.4%, the samples for the direct count were put in cans, sterilized, and counted later. With the lower acidity there were liquifiers which prevented the counting of some plates, so that in the later trials the higher acidity gelatin was used. The count of the molds was not normal, due to the frequent stirrings, which prevented spore formation, besides injuring the hyphae.

4. The Journal of Infectious Diseases. 1909. Vol. 6, p. 393.

The results varied, some pulps giving a much higher initial count than others, but they all agreed in having a comparatively slight increase in the first three hours, the large numbers which one is led to expect not being present until the pulp had stood for at least five hours and under the most favorable conditions; usually it requires a longer time. The plates and the direct count agreeing in the general trend, though the numbers obtained by the two methods varied. In the pulp obtained from the steamed tomatoes, the initial count was much lower in the tomatoes steamed eight minutes, being only 20 per cc in the plates, but the same thing was true of these in that the increase was very slow at first. The figures from all the trials, both raw and steamed pulp, and from the plates and direct counts, show that the theoretical estimation of the increase of organisms from the classic twenty minutes required for reproduction of an organism with the consequent progression, irrespective of the condition of the organism at the start, or its environment, will have to be modified. In the plates all colonies, aside from the molds, were counted as bacteria, but this would not give a very large error, as yeast does not reproduce at the same rate as do bacteria.

The state of comminution of the product determines to a considerable extent the number of organisms which may be counted. The more finely the comminution, the greater the number. Two pulps made from the same

material, one run through an ordinary cyclone and the other through a finishing machine, will show from 50 to 100 per cent more in the latter. Coarse pulp and coarse ketchup may be inferior articles and yet give the better results by the direct method. The effect on the mold is even more marked—filaments and clumps will be torn into many small particles. The total quantity is not increased, but it is distributed more nearly perfectly and thus occurs in more fields.

In work done on meat to determine the technique which should be employed in the bacteriological analysis, comparison was made between shaking the sample and grinding it in a mortar with sand. In the three samples reported, the shaking gave only 3, 12, and 13 per cent, respectively, of those obtained from grinding.^[5]

5. Weinzirl, John and Newton, E. B. American Journal of Public Health. Vol. IV, No. 5.

A finely comminuted pulp was vigorously shaken for definite times and samples taken as quickly as possible after the tenth, fiftieth, one hundredth, and two hundredth times shaken. The results were as follows:

No.	No. Times Shaken.	Bacteria Per c.c.	Yeast and	Mold
			Spores Per 1-60 c.c.	in Per Cent of Fields.
1	10	31,020,000	22	80
2	50	50,040,000	42	76
3	100	84,730,000	106	92
4	200	116,640,000	116	100

In line with this are the results obtained before and after shipping long distances. When the goods have been handled roughly during shipping the count is much higher.

The length of time elapsing after manufacture until the counting is done also has an effect. Pulp put up in the fall will show one count and the same pulp the following season a different count. This difference is not due to any multiplication during storage, but to the fact that the organisms separate from the tissues more readily. The difference made in the counting from this

treatment is not as marked as that produced by the other factors already treated, but is sufficient to cause a change in the count.

It is known that the surface of plants is covered by a variety of bacteria and other fungi that remain dormant under unfavorable conditions, but that these become active when the food which is invariably present is rendered available by access of moisture, either dew or rain, or the rupture of the host, etc. These will vary in numbers with the season, wet or dry, hot or cold, in different sections of the country, and, in the case of the tomato, with the variety of the fruit; whether perfectly smooth or with a slight bloom; whether irregular or regular in shape; and whether slightly green with a firm skin or fully ripe. These are all factors that have an influence and should not be overlooked. Some packers have already learned that by packing tomatoes which are colored, but not really ripe, that the count will be lower, and as such a practice extends, it means the use of poorer material instead of that which is properly developed and with the normal flavor.

- Transcriber's Notes:

- Typographical errors were silently corrected.
- Inconsistent spelling and hyphenation were made consistent only when a predominant form was found in this book.

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